INCH-POUND

MIL-C-48162B (AR)
AMENDMENT 3
12 February 2001
SUPERSEDING
AMENDMENT 2
16 October 1996

MILITARY SPECIFICATION

CARTRIDGE, IGNITION, M702
LOADING, ASSEMBLING AND PACKING

MIL-C-48162B was inactivated before 14 February 1997 for new design.

This amendment forms a part of MIL-C-48162B (AR), dated 14 August 1987, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, Add new military specification as follows:

"MIL-P-60312 - Parts, Molded, Plastic Foam, Polystyrene (For use with ammunition)"

PAGE 2

2.1.2, Product and Packaging Drawing 9287826, delete "Fiberboard" and substitute "Wirebound"

* 2.1.2, Product drawings, delete "INSPECTION EQUIPMENT DRAWING 9278822 Envelope Drawing for Leak Testing of Ignition Cartridge"

PAGE 4

* Add new paragraph 3.11 as follows:

"3.11 Check test for deterioration of ignition cartridge. If the total elapsed time between original acceptance of the ignition cartridge lot and assembly of that lot into the complete round exceeds two years, or if the ignition cartridges have been subjected to adverse conditions for any period of time since previous tests, the ignition cartridges shall be subjected to, and must satisfactory comply with the test in 6.6 before the ignition cartridges are assembled into the completed rounds."

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4.4.2.2, Notes, delete “Test shall be performed on material …. Eight hour production.” And substitute “Test shall be performed on material …. Each production shift.”

* 4.4.2.2, Notes, add second sentence as follows:

“If a Ohaus Moisture Determination Meter is the method of test, the sample shall consist of the pellets required to equal not less than 10 grams.”

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4.4.2.5:

Add critical 2 as follows:

“2. Propellant weight (after loading) … 100% … 3.2 … 4.4.3.1.1/4.5.4.1.”

Major 101, delete in it entirety.

Major 102, delete in its entirety.

* 4.4.2.5:

Add Major 101 as follows:

“101 Propellant weight (after loading) … b … 100% … 3.2 … 4.4.3.1.1/4.5.4.1”

Add note b as follows:

“b. See 4.4.3.1.1 for defect description. Same sample for both inspections.”

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4.4.2.6:

Critical 3, add note (a) after sample size and in notes section which reads:

“(a) This five foot drop test on 25 samples will only be performed on each new lot of primers when they are initially introduced into the M702 production.”

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Add Critical 4 as follows:

"4. Firing plug not loose after assembly … 100% … 3.2 … Manual."

Add Critical 5 as follows:

"5. Propellant weight … ** … 3.2 … 4.4.3.1.2/4.5.4.2."

Special A, delete in its entirety.

Special B, delete in its entirety.

Major 103, delete in its entirety and substitute the following:

"103. Propellant weight … ** … 3.5 … 4.4.3.1.2/4.5.4.2."

Major 104, Delete in its entirety and substitute the following:

"104. Leak test … 60 (b) … 3.5 … 4.5.3."

Add note b as follows:

"b" Test shall be performed on material selected from each days production. If more than 3 failures occur, that days production shall be 100 percent screened."

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4.4.2.6:

Delete Major defect 109 in its entirety.

Add Minor defect 202 as follows:

"202 Marking missing, misleading or unidentifiable … 0.65% … 3.2 … Visual"

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4.4.2.7, Add new major defect 102 and notes a and b as follows:

"102 Moisture content excessive … (a) (b) … 5.1 … (a) (b)

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(a) Requirement and inspection procedures shall be per MIL-P-60312. Two (2) samples shall be randomly selected from a quantity of trays for scheduled production for five (5) consecutive shifts. If one or more samples are defective, the quantity of production represented by the sample shall be rejected. Inspection shall be accomplished within 24 hours of scheduled use of the trays.

(b) Moisture content determination not required when this packing material is stored in an environmentally controlled area maintained at 40 percent maximum relative humidity (RH) at 72 degrees F for at least 24 hours prior to using."

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4.4.2.8, Add new major defect 103 and note (a) as follows:

"103 Moisture content excessive for any packaging component (4 items) … (a) … 5.1 … Gage

(a) Components shall be inspected prior to use every hour during the production shift. If any sample exceeds the requirement, the inspection lot shall be rejected from further processing."

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4.4.2.9, Add new major defect 104 and note (a) as follows:

"104 Environmental conditions inadequate … (a) … 5.1 … Gage

(a) Temperature and humidity shall be checked at the beginning and end of each shift. If conditions are not proper at either time, the quantity of product represented by the conditioning shall be rejected for further processing."

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Add new paragraphs 4.4.2.11, 4.4.2.12, and 4.4.2.13 included:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1 1	Box, Wirebound, Before Packing				9294951
					NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION/CHARACTERISTICS	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/INSPECTION METHOD
Critical:	None defined				
Major					
101.	Inside dimensions incorrect		0.40%	5.1	Gage
102.	Broken, split or missing components		0.40%	5.1	Gage
Minor					
201.	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS MIL-C-48162B (AR)
AMENDMENT 3

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1 2	Packed Wirebound Box, Before Closing				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION/CHARACTERISTICS	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/INSPECTION METHOD
Critical:	None defined				
Major					
101.	Improperly packed		0.40%	5.1	Visual
Minor					
201.	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1 3	Packed Wirebound Box After Closing				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION/CHARACTERISTICS	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/INSPECTION METHOD
Critical:	None defined				
Major					
101.	Wire closing improper		0.40%	5.1	Visual
Minor					
201.	Contents loose		0.65%	5.1	Manual
202.	Marking incomplete, incorrect, or illegible		0.65%	5.1	Visual
203.	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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4.4.3.1.1, Delete in its entirety and substitute the following:

"4.4.3.1.1 Prior to assembly of cap. (See dwg. 9280553). The propellant charge shall be volumetrically dispensed (or tare-weighed) in a container, transferred to the ignition cartridge and then check-weighed 100 percent. The check-weighing shall be as specified in 4.5.4.1. The volumetric dispensing (or initial weighing) and check-weighing shall be performed by different operators on different scales. Weight limits for weighing shall be established within the applicable requirements, depending upon the accuracy and precision of the weighing system to assure compliance with the charge weight requirements. Any charge of bulk propellant failing to meet the applicable requirements shall be classed defective and removed from the lot."

* 4.4.3.1.1, Delete in its entirety and substitute the following:

"4.4.3.1.1 Prior to assembly of cap. (See dwg. 9280553). The propellant charge shall be volumetrically dispensed (or tare-weighed) in a container, transferred to the ignition cartridge and then check-weighed 100 percent. The check-weighing shall be as specified in 4.5.4.1. The volumetric dispensing (or initial weighing) and check-weighing shall be performed by different operators on different scales. Weight limits for weighing shall be established within the applicable requirements, depending upon the accuracy and precision of the weighing system to assure compliance with the charge weight requirements. Any charge of bulk propellant failing to meet the applicable requirements shall be classed defective using criteria for critical and major defects as defined in 4.4.3.1.2 and removed from the lot."

4.4.3.1.2, Delete in its entirety and substitute the following:

"4.4.3.1.2 After assembly of cap. (See dwg. 9280553). Critical defect – more than 8 grains above assessed maximum weight: Critical defect – more than 8 grains below assessed minimum weight: Major defect – 1 to 8 grains above assessed maximum weight: Major defect – 1 top 8 grains below assessed minimum weight. 315 sample ignition cartridge assemblies shall be randomly selected from each lot for this test. If any sample exceeds the assessed maximum weight by more than 8 grains or if two or more samples exceed the assessed maximum weight by 1 to 8 grains, the lot shall be rejected. If any sample is under the assessed minimum weight by more than 8 grains or if two or more samples are under the assessed minimum weight by 1 to 8 grains, the lot shall be rejected. Testing shall be as specified in 4.5.4.2."

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* 4.4.3.1.2 Delete in its entirety and substitute the following:

"4.4.3.1.2 <u>After assembly of cap</u>. (See dwg. 9280553). Critical defect – 8 or more grains above assessed weight. Critical defect – 8 or more grains below assessed weight. Major defect – More than 1 but less than 8 grains above assessed weight. Major defect – More than 1 but less than 8 grains below assessed weight. 315 sample ignition cartridge assemblies shall be randomly selected from each lot for this test. If any sample exceeds the assessed weight by 8 or more grains or if two or more samples exceed the assessed weight by more than 1 grain but less than 8 grains, the lot shall be rejected. If any sample is under the assessed weight by 8 or grains or if two or more samples are under the assessed weight by more than 1 grain but less than 8 grains. The lot shall be rejected. Testing shall be as specified in 4.5.4.2."

4.4.3.1.3, Delete in its entirety.

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4.4.3.2.3, Delete first sentence in its entirety and substitute the following:

"Twenty-five assemblies consisting of randomly chosen primed bodies, heads and firing plugs shall be selected for this test."

4.4.3.2.3, Add to last sentence as follows:

"(Test shall be performed on new primer lots prior to use in M702 production. Acceptable results shall qualify the primer lot for use in M702 production)."

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4.5.1, Delete the test in its entirety and substitute the following:

"The moisture content of the black powder pellets shall be determined in accordance with the procedures specified in MIL-P-223. A sample of 25 black powder pellets shall be randomly selected from each production shift. If any pellet fails to comply with the requirement cited in 3.3 herein, all pellets in the production shift represented by this sample shall be rejected. If loading with propellant has begun, ignition cartridges loaded with the non-conforming black powder pellets shall be rejected."

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* 4.5.1, Delete the second and third sentence in their entirety and substitute the following:

"A sample of 25 black powder pellets or a single sample of not less than 10 grams of pellets shall be randomly selected from each production shift. If any pellet or if the 10 gram sample fails to comply with the requirement cited in 3.3 herein, all pellets in the production shift represented shall be rejected."

4.5.2.3, Delete in its entirety and substitute the following:

"4.5.2.3 Safety drop test. Each of the test assemblies will consist of an inert filled M720 body, and an inert M935 Fuze and an M27 Tailfin assembly. After inserting the sample M702 to be tested, the cartridge total drop weight will be 3.75 pounds +/- 0.05 pounds. The cartridge will be dropped five feet onto a steel plate first in the fin down orientation and then dropped fuze end down. Each cartridge will have the inert M935 fuze and M27 tailfin assembly replaced after each double drop. The drop height shall be measured from the lowest point of the cartridge orientation to the top of the anvil surface. Any test assembly which functions as a result of the drop shall be classed defective. All questionable test assemblies shall be downloaded to verify primer functioning or nonfunctioning. All components used for this test shall not be returned to their respective lots."

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4.5.3, Delete first sentence in its entirety and substitute the following:

"After the ignition cartridges have been assembled and the RTV is applied, the ignition cartridges will be kept in the controlled environmental storage for two (2) hours minimum prior to performing the leak test."

* 4.5.3, Delete "The ignition cartridges shall be tested 100% in accordance with drawing 9278822" and add the following:

"The ignition cartridge shall be subjected to a pressure of 3 + 1 psig for a duration of five (5) seconds minimum. Any ignition cartridge with a leak rate of 1.5 CC per minute or greater shall be rejected and removed from the lot. The test equipment shall be calibrated at the start of each day's operation and prior to production restarts. A restart results from any break in continuous production whether it is due to equipment stoppage/failure or operator breaks. Both accept and reject standards shall be used for each calibration occasion. If the equipment is out of calibration, the ignition cartridge tested since the last successful calibration shall be released after correction of the equipment. Equipment design shall be submitted for approval (See 4.4.4)."

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4.5.4, Delete in its entirety and substitute the following:

"4.5.4 Weight of propellant."

Add paragraph 4.5.4.1 as follows:

"4.5.4.1 Prior to assembly of cap. The ignition cartridge subassembly shall be weighed before and after the propellant is loaded. The weight of the propellant charge shall be determined by subtracting the tare weight of the ignition cartridge subassembly from the gross weight (after loading). The contractor may elect to perform the 100 percent check-weighing operation with automated equipment and mandatory process controls. In such cases, process control plans in accordance with data item description DI-P1604 must be submitted to the technical agency for approval of the automated operations."

Add paragraph 4.5.4.2 as follows:

"4.5.4.2 After assembly of cap. The propellant shall be removed from each ignition cartridge sample, placed on the pan of a precision or analytic balance and weighed. Alternatively, the ignition cartridge assembly may be weighed and reweighed after removal of the propellant: The weight of the propellant charge shall be determined by subtracting the weight of the empty ignition cartridge from the loaded (gross) weight. The weight of the propellant in each sample shall be determined to the nearest 0.01 grain."

* 6.6, Delete in its entirety and substitute the following:

"6.6 Check test for deterioration of ignition cartridge."

* 6.6.1, Delete in its entirety and substitute the following:

"6.6.1 Functioning. The requirement shall be as specified in 3.4.2. A sample of 150 each ignition cartridges shall be used. If one or more primers fail to comply with the specified requirement, the lot shall be rejected. The test shall be performed as specified in 4.5.2.2."

* Add 6.6.2 as follows:

"6.6.2 Moisture content of black powder pellet. Twenty five samples of black powder pellets shall be randomly selected from the ignition cartridge lot. During disassembly, the ignition

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cartridge metal parts shall be visually examined for evidence of corrosion or discoloration. The requirement for black powder moisture content is specified in 3.3 and test shall be performed as specified in 4.5.1. The lost shall be rejected if the moisture content fails to comply with the requirement or evidence of corrosion or discoloration exists on metal parts."

* Add 6.6.3 as follows:

"6.6.3 Nonfunctioning. The requirement shall be as specified in 3.4.1. A sample of 200 each ignition cartridges shall be used. If one or more primers fails to comply with the specified requirement, the lot shall be rejected. The test shall be performed as specified in 4.5.2.1."

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